

Stellar Feedback as Holographic Information Regulation

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Abstract

We develop an operational model of stellar feedback within the holographic framework of Ontological Resolution Theory (ORT), demonstrating that stellar evolution is not merely a local thermodynamic process but an active component of cosmological information regulation. Starting from three principles—(1) gravitational pumping creating free-energy gradients, (2) quantum resistance via Planck’s constant $\hbar$ limiting information processing rates, and (3) holographic relaxation through stellar radiation—we derive a closed informational cycle connecting stellar astrophysics to the finite capacity of the cosmological horizon ($\mathcal{R}_{\text{cosm}} \sim 2.9 \times 10^{122}$ bits). We formulate an effective coarse-grained horizon dynamics via a screened Poisson-like equation on S^2 and derive the diffusion coefficient from light-crossing constraints consistent with the Margolus–Levitin bound. Quantitative predictions include entropy production rates ($\dot{S}_{\text{rad}}/\dot{S}_{\text{star}} \sim 10^5\text{--}10^6$ for the Sun), total information processing ($I_{\odot} \sim 10^{78}$ bits, comparable to a $\sim 3M_{\odot}$ black hole), and relaxation timescales ($\tau_{\text{relax}} \sim 10^8\text{--}10^9$ yr for galaxy cluster scales). This framework unifies stellar thermodynamics with holographic information bounds, suggesting that stellar radiation provides negative feedback preventing permanent informational saturation of the cosmological horizon.

Keywords: stellar evolution, holographic principle, information thermodynamics, cosmological horizon, Planck’s constant, entropy production

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1 Introduction

[...previous introduction text unchanged...]

2 Horizon Dynamics and Effective Poisson Equation

2.1 Setup

Consider the cosmological event horizon as a two-sphere S^2 of radius

$$R_H = \frac{c}{H_0}, \tag{1}$$

where we adopt $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1} = 2.27 \times 10^{-18} \text{ s}^{-1}$, giving

$$R_H \approx 1.32 \times 10^{26} \text{ m}. \quad (2)$$

The total informational capacity is

$$\mathcal{R}_{\text{cosm}} = \frac{A_H}{4\ell_P^2 \ln 2} \approx 2.9 \times 10^{122} \text{ bits}, \quad (3)$$

where $A_H = 4\pi R_H^2 \approx 2.19 \times 10^{53} \text{ m}^2$ and $\ell_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35} \text{ m}$.

We introduce:

- $\Phi(\Omega, t)$: effective gravitational potential on S^2 (dimension: m^2/s^2)
- $\Sigma(\Omega, t) = \int_0^{R_H} \rho(r, \Omega, t) dr$: projected column density (dimension: kg/m^2)

2.2 Effective Horizon Equation

We postulate an *effective coarse-grained* equation relating Φ and Σ on S^2 :

$$\Delta_{S^2} \Phi + \frac{2}{R_H^2} \Phi = -\frac{4\pi G}{R_H} \Sigma, \quad (4)$$

where Δ_{S^2} is the Laplace–Beltrami operator on the two-sphere of radius R_H .

Dimensional check:

$$[\Delta_{S^2} \Phi] = \frac{[\Phi]}{R_H^2} = \frac{\text{m}^2/\text{s}^2}{\text{m}^2} = \text{s}^{-2}, \quad (5)$$

$$\left[\frac{2\Phi}{R_H^2} \right] = \text{s}^{-2}, \quad (6)$$

$$\left[\frac{4\pi G \Sigma}{R_H} \right] = \frac{\text{m}^3}{\text{kg s}^2} \times \frac{\text{kg}}{\text{m}^2} \times \frac{1}{\text{m}} = \text{s}^{-2}. \quad (7)$$

Dimensions match correctly.

Motivation: This is not an *exact* projection from 3D general relativity but rather a phenomenological *effective 2D model* motivated by:

1. The screened nature of gravitational interactions on curved surfaces (the $+2\Phi/R_H^2$ term arises from the intrinsic curvature $K = 1/R_H^2$ of S^2 , ensuring consistency with flat-space limits as discussed in [10]).
2. Spherical harmonic decomposition: for modes with degree $\ell \gg 1$, Δ_{S^2} eigenvalues are $-\ell(\ell+1)/R_H^2$, and the additional $1/R_H$ factor in the source term reflects the *radial integration* implicit in defining Σ as a column density.
3. This structure is analogous to the membrane paradigm for black hole horizons [11], adapted to cosmological scales.

We emphasize that Eq. (4) should be understood as a *coarse-grained effective description*, not a rigorous derivation from first principles. Its justification lies in:

- Correct dimensional scaling,
- Recovery of expected limits (black holes, flat space),
- Consistency with holographic bounds (see below).

2.3 Connection to ORT: Phenomenological Ansatz

The potential depth quantifies informational cost. We propose a phenomenological ansatz relating Φ to local informational allocation:

$$\mathcal{R}_{\text{local}}(\Omega) = \alpha \frac{|\Phi(\Omega)|}{c^2} \times \frac{A_H}{\ell_P^2}, \quad (8)$$

where α is a dimensionless proportionality constant of order unity.

Justification: This is an *effective correspondence*, not derived from ORT axioms, chosen such that:

1. **Dimensionality:** Φ/c^2 is dimensionless, and A_H/ℓ_P^2 has dimensions of bits (via holographic entropy).
2. **Black hole limit:** At the Schwarzschild radius $r_S = 2GM/c^2$, the Newtonian potential is $\Phi(r_S) \approx -c^2/2$. Substituting into Eq. (8) and replacing $A_H \rightarrow A_{\text{BH}} = 4\pi r_S^2$:

$$\mathcal{R}_{\text{BH}} \sim \alpha \frac{c^2/2}{c^2} \times \frac{4\pi r_S^2}{\ell_P^2} = 2\pi\alpha \frac{r_S^2}{\ell_P^2}. \quad (9)$$

The Bekenstein–Hawking entropy in bits is

$$I_{\text{BH}} = \frac{A_{\text{BH}}}{4\ell_P^2 \ln 2} = \frac{4\pi r_S^2}{4\ell_P^2 \ln 2} = \frac{\pi}{\ln 2} \frac{r_S^2}{\ell_P^2}. \quad (10)$$

Matching these expressions gives

$$2\pi\alpha = \frac{\pi}{\ln 2} \quad \Rightarrow \quad \alpha = \frac{1}{2 \ln 2} \approx 0.72. \quad (11)$$

However, given that this is a phenomenological model, we do not insist on this precise value; α is an $\mathcal{O}(1)$ constant chosen to reproduce the black hole limit up to factors of order unity.

3. **Saturation criterion:** When $\mathcal{R}_{\text{local}}$ approaches the holographic bound locally, event horizons form (Section 6 of Ref. [1]).

This ansatz provides an operational bridge between gravitational potentials and information allocation, but we acknowledge it as a working hypothesis requiring further theoretical justification.

2.4 Dynamical Evolution

To describe time-dependent feedback, we supplement Eq. (4) with:

$$\frac{\partial \Phi}{\partial t} = -c^2 \nabla_{S^2} \cdot \mathbf{J} - \frac{4\pi G}{R_H} \frac{\partial \Sigma}{\partial t} + \gamma \Delta_{S^2} \Phi, \quad (12)$$

where:

- $\mathbf{J}(\Omega, t)$: effective information flux vector on S^2 (has effective dimension of a tangential velocity field on the sphere)

- γ : holographic diffusion coefficient (dimension: m^2/s)

Remark 1 (Effective nature of $\mathbf{J}$). The field $\mathbf{J}$ represents the tangential redistribution of holographic degrees of freedom on S^2 . It is not a direct physical energy or mass flux but rather an *effective field* in our coarse-grained model. From our work on information flow dynamics [2], it is related to gradients in informational pressure:

$$\mathbf{J} \sim -\mu \nabla_{S^2} P_I, \quad (13)$$

where $P_I \propto \rho_I \dot{I}$ and μ is informational mobility. Stellar radiation creates gradients in ρ_I , driving these flows. Given the effective nature of our horizon dynamics, we do not attempt a rigorous dimensional analysis of $\mathbf{J}$ but note it has dimensions consistent with a velocity field on the sphere.

Derivation of γ : For patches separated by angle $\Delta\theta$, the light-crossing time is

$$\tau_{\text{cross}} = \frac{R_H \Delta\theta}{c}. \quad (14)$$

The diffusion coefficient scales as

$$\gamma \sim \frac{(\text{distance})^2}{\text{time}} \sim \frac{(R_H \Delta\theta)^2}{\tau_{\text{cross}}} = c R_H \Delta\theta. \quad (15)$$

For spherical harmonic degree ℓ (where $\Delta\theta \sim 1/\ell$):

$$\gamma(\ell) \sim \frac{c R_H}{\ell}. \quad (16)$$

This is a **lower bound** from causality: information cannot propagate faster than light over distance $\sim R_H/\ell$. The Margolus–Levitin theorem [12] provides additional support:

$$\dot{N}_{\text{ops}} \leq \frac{2E}{\pi\hbar}, \quad (17)$$

which translates to a maximum information transfer rate $\sim E/\hbar$ per degree of freedom, consistent with γ being proportional to $c \times \text{length}$.

3 Three Principles of Informational Thermodynamics

[...Principles 1 and 2 unchanged...]

Principle 1 (Holographic Relaxation). *Radiative outflows reduce $\Sigma(\Omega, t)$, driving $\partial\Phi/\partial t > 0$ via Eq. (12). The diffusion term $\gamma\Delta_{S^2}\Phi$ prevents permanent informational saturation.*

The relaxation timescale for a perturbation of spherical harmonic degree ℓ is

$$\tau_{\text{relax}}(\ell) \sim \frac{R_H^2}{\gamma(\ell)\ell(\ell+1)}. \quad (18)$$

Substituting $\gamma(\ell) \sim cR_H/\ell$ from Eq. (16):

$$\tau_{\text{relax}}(\ell) \sim \frac{R_H^2}{(cR_H/\ell) \cdot \ell^2} = \frac{R_H}{c\ell}. \quad (19)$$

Numerical estimates: With $R_H \approx 1.32 \times 10^{26} \text{ m}$ and $c = 3 \times 10^8 \text{ m/s}$:

$$\frac{R_H}{c} \approx 4.4 \times 10^{17} \text{ s} \approx 1.4 \times 10^{10} \text{ yr.} \quad (20)$$

Thus:

- For $\ell = 10$ (physical scale $L \sim 2\pi R_H/\ell \approx 80 \text{ Mpc}$, supercluster scales):

$$\tau_{\text{relax}}(\ell = 10) \sim \frac{1.4 \times 10^{10}}{10} \text{ yr} \approx 1.4 \times 10^9 \text{ yr.} \quad (21)$$

- For $\ell = 100$ (physical scale $L \sim 8 \text{ Mpc}$, galaxy cluster scales):

$$\tau_{\text{relax}}(\ell = 100) \sim 1.4 \times 10^8 \text{ yr.} \quad (22)$$

These timescales are shorter than the Hubble time $H_0^{-1} \approx 1.4 \times 10^{10} \text{ yr}$ but still comparable to large-scale structure formation timescales, enabling bidirectional coupling between local stellar processes and global horizon dynamics on cosmologically relevant scales.

[...cycle diagram...]

4 Quantitative Predictions and Tests

4.1 Solar Entropy Production

For the Sun ($L_\odot = 3.83 \times 10^{26} \text{ W}$, $T_{\text{eff}} = 5778 \text{ K}$):

$$\begin{aligned} \dot{S}_{\text{rad},\odot} &= \frac{L_\odot}{T_{\text{eff}}} \approx 6.6 \times 10^{22} \text{ J}/(\text{K} \cdot \text{s}) \\ &\approx \frac{6.6 \times 10^{22}}{1.38 \times 10^{-23}} k_B/\text{s} \approx 4.8 \times 10^{45} k_B/\text{s}. \end{aligned} \quad (23)$$

The internal stellar entropy change rate can be estimated as:

$$\dot{S}_{\text{star},\odot} \sim \frac{S_{\text{star}}}{\tau_\odot} \sim \frac{k_B \times 10^{57}}{3 \times 10^{17} \text{ s}} \sim 3 \times 10^{39} k_B/\text{s}. \quad (24)$$

Ratio:

$$\frac{\dot{S}_{\text{rad}}}{\dot{S}_{\text{star}}} \sim \frac{4.8 \times 10^{45}}{3 \times 10^{39}} \sim 1.6 \times 10^6, \quad (25)$$

confirming that radiated entropy vastly exceeds internal changes by $\sim 10^5$ – 10^6 , consistent with the Second Law globally even as local stellar entropy may decrease during contraction phases.

4.2 Information Processing and Holographic Consistency

From the Margolus–Levitin bound, the maximum information processing rate for the Sun is

$$\dot{I}_{\odot,\text{max}} \approx \frac{L_\odot}{\hbar \ln 2} \approx \frac{3.83 \times 10^{26}}{1.054 \times 10^{-34} \times 0.693} \approx 5.2 \times 10^{60} \text{ bits/s.} \quad (26)$$

Over the main-sequence lifetime $\tau_{\odot} \approx 10^{10}$ yr $\approx 3.15 \times 10^{17}$ s:

$$I_{\odot, \text{total}} \lesssim 5.2 \times 10^{60} \times 3.15 \times 10^{17} \approx 1.6 \times 10^{78} \text{ bits.} \quad (27)$$

This is comparable to the Bekenstein–Hawking entropy of a $\sim 3M_{\odot}$ black hole:

$$S_{\text{BH}}(3M_{\odot}) = \frac{A_{\text{BH}}}{4\ell_P^2 \ln 2} \approx \frac{\pi r_S^2}{\ell_P^2 \ln 2} \approx 1.2 \times 10^{78} \text{ bits.} \quad (28)$$

Interpretation: This numerical coincidence suggests that typical stellar lifetimes and luminosities are such that the *total potential* information processing (if saturating the Margolus–Levitin bound) is of the same order as the maximum horizon information for their remnant mass. While actual stellar processes operate far below this limit (due to opacity, reaction rates, etc.), the fact that integrating the theoretical maximum over $\tau_{\odot}$ gives $\sim S_{\text{BH}}$ hints at a deeper connection between stellar evolution and holographic information bounds.

This does *not* imply that stars literally “saturate” their information capacity in practice, but rather that the *scale* of stellar information processing is naturally set by the same holographic parameters (ℓ_P , G , $\hbar$) that govern black hole entropy.

4.3 Global Entropy Budget

[...rest unchanged...]

5 Conclusion and Future Directions

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